

Course Syllabus

1. Course Number: 2110403
2. Course Credit: 3 (3-0-6)
3. Course Title: Introduction to Data Science and Data Engineering
4. Department: Computer Engineering
5. Semester: First Semester
6. Academic Year: 2026
7. Instructor: Prof. Peerapon Vateekul, Ph.D.
Asst. Prof. Natawut Nupairoj, Ph.D.
Assoc. Prof. Veera Muangsin, Ph.D.
8. Condition: -
9. Status: Required
10. Curriculum: Computer Engineering & Digital Technology (CEDT)
11. Degree: B.Eng.

12. Course Description

Data Science (DS) is the study of the discovery of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases, and other branches of computer science along with a good understanding of the craft of problem. Data Engineering (DE) is the study of how to engineer or process data, i.e., data cleansing, data storing, etc. There are three main parts in this course:

- Data engineering: Data exploration & preparation
- Data science: Machine Learning techniques
- Data visualization: Storytelling via data

13. Course Outline

13.1. Learning Objectives

- Describe what DS and DE are and the skill sets needed
- To be able to explore and understand collected data
- To be able to analyze data by apply traditional machine learning techniques
- To be able to visualize data in relation to spatial and temporal points of views

13.2. Learning Contents

- Sections 51, 52, 53, and 54 on the 3rd floor of the Chula Phat Building.
- 2nd Block – Weeks 10 to 18 (October 6 – December 2, 2025)
- Class schedule: Tue 9AM – 12PM*, Wed 9AM – 12PM & 1PM – 4PM
 - There are two lecture sessions each week.
 - The Wednesday afternoon session is reserved for practice.
- *** Please bring your laptop to the class ***

W	#	DoW	Date		Contents	Instructor	Module
10	1	Tue	6-Oct-26	9-12AM	Introduction + Basic Python	Aj.Peerapon	DS1
10	2	Wed	7-Oct-26	9-12AM	TA session	Aj.Peerapon	
				1-4PM	Data Prep	Aj.Peerapon	DS2
11	3	Tue	13-Oct-26	9-12AM	TA session	Aj.Peerapon	
11	4	Wed	14-Oct-26	9-12AM	Traditional ML (1)	Aj.Peerapon	DS3
				1-4PM	Traditional ML (2)	Aj.Peerapon	DS4
12	5	Tue	20-Oct-26	9-12AM	TA session	Aj.Peerapon	
12	6	Wed	21-Oct-26	9-12AM	Traditional ML (3)	Aj.Peerapon	DS5
				1-4PM	Traditional ML (4)	Aj.Peerapon	DS6
13	7	Tue	27-Oct-26	9-12AM	Midterm Practice (On-site, required)	Aj.Peerapon	
13	8	Wed	28-Oct-26	9-12AM	Deep Learning	Aj.Peerapon	DS7
				1-4PM	Advanced topics (Prompt Engineering, RAG, Agentic AI)	Aj.Peerapon	DS8
					Midterm exam on Sun 1 Nov 2026 for 2 rounds (8.30 - 11.30 & 13.00 - 16.00)		
14	8	Tue	3-Nov-26	9-12AM	Intro to Viz + BI + Storytelling	Aj.Veera	VIZ1
14	9	Wed	4-Nov-26	9-12AM	Big data architecture + data storage	Aj.Natawut	DE1
				1-4PM	TA session	Aj.Natawut	
15	10	Tue	10-Nov-26	9-12AM	Web scraping	Aj.Natawut	DE2
15	11	Wed	11-Nov-26	9-12AM	Data ingestion	Aj.Natawut	DE3
				1-4PM	TA session	Aj.Natawut	
16	12	Tue	17-Nov-26	9-12AM	Big data processing (Spark)	Aj.Natawut	DE4
16	13	Wed	18-Nov-26	9-12AM	Intro to Viz + BI + Storytelling	Aj.Veera	VIZ1
				1-4PM	TA session		
17	14	Tue	24-Nov-26	9-12AM	Python visualization	Aj.Veera	VIZ2
17	15	Wed	25-Nov-26	9-12AM	Graph analysis & spatial analysis	Aj.Veera	VIZ3
				1-4PM	Final Practice (On-site, required)	All prof.	
18	16	Tue	1-Dec-26		No Class		
18		Wed	2-Dec-26		Final Exam on Wed 2 Dec 2026 for 2 rounds (8.30 - 11.30 & 13.00 - 16.00)		

* There can be up to two guest speakers in the class.

13.3. Method: Lecture and Lab

13.4. Learning Media: PowerPoint presentation, Zoom

13.5. Evaluation

- Attendance 5%
- Midterm Exam 30% (Lab Test)
- Final Exam 30% (Lab Test)
- Assignment 20%
- Project 15%

14. Reading List

14.1. Required Text: N/A

14.2. Electronic Media or Websites:

15. LMS

15.1. MyCourseVille: <https://mycourseville.com/?q=courseville/course/86514>

15.2. Discord: <https://discord.gg/na8GkXVp2B>

15.3. Github: https://github.com/pvateekul/2110403_DSDE-CEDT_2026s1

15.4. Grader: <https://dsde.nattee.net/>